

# Python in One Line

EEC 3150

Trevecca Nazarene University

# Outline

1 Inline If Statements

2 List Comprehension

# Inline If Statements

# Inline If Statements

## Inline if statements:

You can write simple if statements in one line in the following way:

```
ans = x if condition else y
```

Examples:

- ```
x = 10  
y = 0 if x%2==0 else 1
```
- ```
n = 3  
x = 'a' if n==0 else 'b' if n==1 else 'c'
```

You can't do inline `elif` but you can achieve the same result by nesting inline ifs

# List Comprehension

# List Comprehension

## List Comprehension:

You can create lists and dictionaries in one line using comprehension:

```
myList = [ expr for i in iterable ]
```

```
myDict = [ key:value for i in iterable ]
```

Examples:

- `x = [ i for i in range(10) ]` (integers 0-9)
- `x = [ i for i in range(10) if i%2==0 ]` (even integers 0-9)
- `x = [ y[i] for i in range(len(y)) if y[i]>thresh ]` (filter another list)
- `x = [ [0 for j in range(5)] for i in range(5) ]` (5x5 of 0's)